Kathmandu University School of Education

Department of STEAM Education

Bachelor of Technical Education (B Tech Ed) Program

Information Technology (IT)

In Association with School of Engineering, KU

Course Curriculum

Semester IV

2022

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Overview of the B Tech Ed Program

Technical and Vocational Education and Training (TVET) has been one of the prominent sectors globally for a long. Dewey (1916) saw TVET as a tool for education reform in modernizing society. This has been equally vibrant in the present context as scholars (e.g., Maclean & Wilson, 2009) argue that TVET is an education and training that prepares an individual for gainful employment. TVET programs equip young people with the skills, knowledge, and competencies required to enter a particular type of professional career (OECD 2017). Hence, the role of TVET in preparing skilled human resources and enabling them to transition into the career path for gainful employment in specific and the country's economic development in a broad is inevitable. The formal TVET began in Nepal only around mid of the 20th century. Nevertheless, occupational skills have been transformed from generation to generation for a long (Ministry of Education [MoE], 2012). The establishment of the Council for Technical Education and Vocational Training (CTEVT), the governing body of TVET in 1989, was a substantive effort toward its development and expansion (CTEVT, 2019). Additionally, at present, the Centre for Education and Human Resource Development (CEHRD) offering technical diplomas in 284 technical stream schools throughout the country (GoN, 2018) and universities (such as Kathmandu University, Manmohan Technical University, etc.) run bachelor and master in TVET and other 11 federal ministries offer TVET related formal and non-formal programs (MoEST, 2019). The CTEVT, which is in charge of formulating TVET-relevant policies, developing standards for programs and curriculums, coordination, accreditation, monitoring, and supervision (CTEVT, 2019) largely shares the TVET related activities. There are 1131 CTEVT affiliated and constituent institutions with roughly the average capacity of 70,000 per annual operate formal TVET programs (CTEVT, 2020). Further expansion of TVET institutions in the country is yet to be expected soon since the government has a provision of establishing at least one technical school in all local levels.

Bachelor of Technical Education is a program to develop instructors and teachers in the field of Technical and Vocational Education and Training (TVET) with a view to fulfil the gap of acute shortage of instructors and TEVT professionals in the country which aspires to promote technical and vocational education two-third of the educational system and processes. The program has envisaged that a mix of two or more of the work-based learning, activity-based instruction, on the job training, project-based learning and problem-based learning be used to deliver the program. Because of the ‘practical focus’ of the program, the assessment has been proposed to address the needs of experiential learning.

The 136-credit program has been structured four types of the courses, such as Educational Core (18), Vocational Specialization (80), Vocational Pedagogy (20), and Educational Research and Internship (18) with a view to inculcate both pedagogical and vocational skills. The vocational specializations have been offered in a manner that they can be utilize hands-on activities as well. The program begins with the foundational courses in general education, vocational pedagogy and the area of specialization. In the later part of the program, OJT/WBL and forms of experiential learning shall be introduced. The culmination of the program will be internships in their specializations and a research report that addresses the problem of vocational education.

B Tech Ed in Information Technology

Bachelor of Technical Education is a 4-year (8 semesters) program to develop instructors, teachers, professionals in the field of Technical and Vocational Education and Training (TVET) with a view to fulfil the gap of acute shortage of instructors and TEVT professionals in the country which aspires to promote technical and vocational education two-third of the educational system and processes. The program has envisaged that a mix of two or more of the work-based learning, activity-based instruction, on the job training, project-based learning and problem-based learning be used to deliver the program. Because of the 'practical focus' of the program, the assessment has been proposed to address the needs of experiential learning. The B Tech Ed in IT program aims at providing technical, vocational, and educational exposures for students who can become IT professionals, IT Instructors, TVET educators, TVET curriculum developers, etc. The occupational/specialization courses emphasize the development of core IT skills in the field of computer science and technology innovation. The B Tech Ed in IT graduates will be eligible for doing master's program in IT, TVET, Education, Social Sciences, etc.

Major Objectives of the B Tech Ed in IT Program

- Demonstrate Level 6 competencies envisaged by the National Vocational Qualification Framework (NVQF) of Nepal,
- Exhibit comprehensive, meaningful, and coherent knowledge and skills in the field of technology and computer science
- Apply occupational specialization knowledge and skills in solving problems occurred in their contexts of work,
- Apply vocational pedagogical knowledge and skills in technical and vocational instruction of Information Technology,
- Show scholarly literacy, communication, quantitative reasoning, critical thinking, learning skills needed for advanced and work-based learning,
- Exemplify a deep and principled understanding of the technical and vocational learning processes and the role of the instructors in facilitating these processes in the students,
- Apply a wide range of teaching process skills including curriculum development, lesson planning, materials development, assessment, and pedagogical approaches, and
- Enable students to become resilient and lifelong learners by harnessing internal and external assets as TVET professionals.

Career Prospects

- TVET Instructors and Educators
- TVET Curriculum Developers
- IT Professionals (Software and Apps developer, programmer, Network Engineer, Web Developer, IT Admin, IT Engineer, etc.)
- Eligible for doing master's program in IT, TVET, Education, Social Sciences, etc.

Entry Requirements

The minimum prerequisite for this program is the completion of Level 5 education (as per National Vocational Qualification Framework) in relevant vocational specializations. The details are-

- Minimum D+ grade in each subject of grade 12 with CGPA 1.8 or more

  Or
- A minimum score of second division marks in 10+2, PCL or equivalent in any disciplines

Core Courses

EDUC 100: NEPL 200 General Nepali

Course: General Nepali **Code:** NEPL 200
Credit Hours: 3 (2 Th + 1 PR) **Nature:** Theory and Practical

विषयको परिचय

परिचय :

यो पाठ्यक्रम काठमाडौ विश्वविद्यालय, स्कुल अफ एजुकेसन अन्तर्गत बि टेक एड (आइ टि तथा सिभिल) अध्ययन गर्ने विद्यार्थीहरुमा नेपाली भाषामा बोध, अभिव्यक्ति र रचना शिल्पको विकास गर्न तयार पारिएको हो । यसमा नेपाली भाषाका भाषातत्व, शब्दभण्डार, सङ्कथन रचना, विविध क्षेत्रका नेपाली गद्यांशको पठनबोध, व्यबहारिक लेखनका साथै निर्धारित साहित्यिक कृतिहरुको पठन, आस्वादन र अभिव्यक्ति क्षमताको विकास गर्ने पाठ्य सामग्रीहरु समावेश गरिएका छन् ।

उद्देश्यहरु :

यो पाठ्यांशको अध्यापनपश्चात विद्यार्थीहरु निम्नलिखित कुराहरुमा सक्षम हुनेछन् :

क) कथ्य र लेख्य नेपालीको भिन्नता पहिल्याई मानक र स्तरीय रुपको प्रयोग गर्न,

ख) विविध क्षेत्रमा प्रयुक्त नेपाली शब्दहरुको स्रोत र संरचना पहिचान गरी सन्दर्भपूर्ण वाक्यमा प्रयोग गर्न,

ग) निर्धारित व्यावहारिक पत्रहरु तयार गर्न,

घ) तालिका, चित्राकृति (डायग्राम) र रेखाचित्र (ग्राफ)मा रहेको सूचनालाई अनुच्छेदमा रुपान्तरण गर्न तथा अनुच्छेदमा रहेको सूचनालाई तालिका, चित्राकृति र रेखाचित्रमा रुपान्तरण गर्न,

ङ) सम्बद्ध गद्यांशको बुँदा टिपोट र संक्षेपीकरण गर्न,

च) निर्धारित साहित्यिक कृतिहरुको आस्वादन गर्न र निर्धारित ढाँचामा ती कृतिहरुको समीक्षा तयार गर्न,

विशिष्ट उद्देश्य तथा पाठ्यविषय

विशिष्ट उद्देश्य	पाठ्यविषय
• शब्दहरुको अक्षरीकरण गर्न • वर्णविन्यासको समुचित प्रयोग गर्न ।	**एकाइ एक : अक्षरीकरण र वर्णविन्यास (४ घण्टा)** क) शब्दहरुको अक्षरीकरण ख) नेपाली वर्णविन्यास र भाषिक चिन्हहरु
• शब्दहरुको स्रोत र निर्माण प्रक्रिया देखाउन ।	**एकाइ दुई : शब्द स्रोत र निर्माण प्रक्रिया (२ घण्टा)** क) शब्द स्रोत : तत्सम, तद्भव, आगन्तुक र ठेट/झर्रा ख) शब्द निर्माण : सर्ग, समास, सन्धि र द्वित्व
• निर्देशित वाक्य रचना गर्न । • स्वतन्त्र वाक्य रचना गर्न(वर्णनात्मक)।	**एकाइ तीन : वाक्यतत्वपरक रचना (४ घण्टा)** क) स्वतन्त्र वाक्यरचना (व्याकरणात्मक कोटीका आधारमा वाक्यरचना) ख) निर्देशित वाक्यरचना (वाक्यान्तरण)
• नेपाली साहित्यका विभिन्न क्षेत्रसँग सम्बन्धित दृष्टांश तथा अदृष्टांश पाठको अध्ययन गरी बोध प्रश्नको समुचित उत्तर दिन ।	**एकाइ चार : पठनबोध .(६ घण्टा)** क) साहित्य, भूगोल, सूचना र सञ्चार, विज्ञान र प्रविधि, दर्शन मनोविज्ञान, कला संस्कृति, जीव र वनस्पति, कानुन दर्शन आदि विविध क्षेत्रसँग सम्बन्धित सामान्य तथा विशिष्ट प्रकृतिका दृष्टांश तथा अदृष्टांश पाठमा आधारित तथ्यात्मक, प्रतिक्रियामूलक, अनुमानात्मक, संरचनात्मक, निष्कर्षात्मक तथा बहुवैकल्पिक बोध प्रश्नहरुको अभ्यास ।

• विविध प्रकारका अनुच्छेदबाट बुँदाटिपोट र सङ्क्षेपीकरण गर्न ।	**एकाइ पाँच : बुँदाटिपोट र सङ्क्षेपीकरण (२ घण्टा)** क) बुँदाटिपोट ख) सङ्क्षेपीकरण
• सूचना, भरपाई, बधाई, शुभकामना, विज्ञापन, श्रद्धाञ्जली, व्यक्तिवृत्त, सम्पादकलाई चिठी, निर्णय र प्रस्ताव तयार गर्न ।	**एकाइ छ : व्यावहारिक लेखन (१० घण्टा)** क) सूचना, भरपाई, विज्ञापन, सम्पादकलाई चिठी ख) बधाई, श्रद्धाञ्जली, शुभकामना ग) निर्णय, व्यक्तिवृत्त घ) प्रस्ताव लेखन
• सभासमारोह, महोत्सव, यात्रा, घटना, अवस्था आदि विषयमा निरीक्षण प्रतिवेदन तयार गर्न ।	**एकाइ सात : प्रतिवेदन लेखन (५ घण्टा)** क) सभासमारोह, महोत्सव, यात्रा, घटना, अवस्था आदि विषयमा निरीक्षण प्रतिवेदन लेखन ।
• तालिका, चित्राकृति (डायग्राम) र रेखाचित्र (ग्राफ)मा रहेको सूचनालाई अनुच्छेदमा रुपान्तरण गर्न तथा अनुच्छेदमा रहेको सूचनालाई तालिका, चित्राकृति र रेखाचित्रमा रुपान्तरण गर्न,	**एकाइ आठ : सूचनाको रुपान्तरण (४ घण्टा)** क) तालिकालाई अनुच्छेदमा र अनुच्छेदको सूचनालाई तालिकामा रुपान्तरण ख) चित्राकृति (डायग्राम) लाई अनुच्छेदमा र उपलब्ध सूचनालाई चित्राकृतिमा रुपान्तरण ग) रेखाचित्र (ग्राफ) लाई अनुच्छेदमा र सूचनालाई रेखाचित्रमा रुपान्तरण
• कविता, कथा, उपन्यास र निबन्धको अध्ययन गरी तिनको आस्वादन गर्न र सङ्क्षिप्त समीक्षा गर्न ।	**एकाइ नौ : साहित्यिक कृतिको अध्ययन तथा समीक्षा (८ घण्टा)** क) कविता • हामी (भूपि शेरचन) • किन चाहियो (मोतीराम भट्ट) ख) कथा • मैले नजन्माएको छोरो (पारिजात) • सङ्ग्रामबहादुर सार्की (खगेन्द्र संग्रौला) ग) निबन्धः • डरको छायाँमा रोबोटको जन्म (कृष्णचन्द्रसिंह प्रधान) घ) उपन्यास • निर्वाण (घनश्याम खड्का) कृति समीक्षाको ढाँचाः • कृतिको नाम • कृतिकारको परिचय • कृतिको मुख्य विषय • कृतिको महत्व • कृतिको भाषाशैली • कृतिको कमी, कमजोरी र सुझाव • कृतिले आफूलाई पारेको प्रभाव

शिक्षण सिकाइ विधि : व्याख्यान, छलफल, समूहकार्य, परियोजना कार्य, प्रस्तुतीकरण ।

मूल्याङ्कन पद्धति :

क) आन्तरिक परीक्षा (५०)

- प्रथम (१०)

- द्वितीय (२०)
- परियोजना कार्य, प्रस्तुतीकरण, कक्षा सहभागिता (२०)

ख) शत्रान्त परीक्षा (५०)

- वस्तुगत (१०)
- विषयगत (४०)

<u>शत्रान्त परीक्षाका लागि मूल्याङ्कन योजना:</u>

- **बहुवैकल्पिक प्रश्न(MCQ): - १० × १ अङ्क**
- **छोटो उत्तरात्मक प्रश्न(SAQ): – जम्मा ७ प्रश्न × ४ अङ्क (कुनै ६ वटा मात्र)**
- **लामो उत्तरात्मक प्रश्न(LAQ) : – जम्मा ३ प्रश्न × ८ अङ्क (कुनै २ वटा मात्र)**

एकाइ	MCQ	SAQ	LAQ
एकाइ १ र २	१	१	×
एकाइ ३	१	१	×
एकाइ ४	×	१	×
एकाइ ५	१	१	×
एकाइ ६	१	२	×
एकाइ ७	२	×	१
एकाइ ८	२	१	×
एकाइ ९	२	×	२
जम्मा	१०	७	३

<u>Note</u>:

MCQ = Multiple Choice Questions

SAQ = Short Answer Questions

LAQ = Long Answer Questions

सन्दर्भ सामग्री :

- मोहनराज शर्मा, *शब्द रचना र वर्णविन्यास,* नवीन प्रकाशन, काठमाडौं
- भूपी शेरचन, *घुम्ने मेचमाथि अन्धो मान्छे,* साझा प्रकाशन, २०५५
- देवीप्रसाद गौतम र कृष्णप्रसाद घिमिरे, *आधुनिक नेपाली कथा, भाग ३,* साझा प्रकाशन, २०६८
- महादेव अवस्थी, *नेपाली कथा, भाग २,* साझा प्रकाशन, २०५५
- कृष्णचन्द्रसिंह प्रधान, *पाइला आगतमा टेकेर,* साझा प्रकाशन, २०५५
- घनश्याम खड्का, *निर्वाण,* अल्केमी पब्लिकेसन, २०७३

Vocational Pedagogical Course

VPED 220 Instructional Skills (II)

Course: VPED 220 Instructional Skills (II)	**Code:** VPED 220
Credit Hours: 2 (1 TH + 1 PR)	**Nature:** Theory and Practical

Course Description

This course is designed to provide students with a foundational aspect of instructional design, learning theories, classroom management, and assessment tools. Students will gain insights into the conceptual framework of instruction and its components – models, strategies, methods, and skills to enhance teaching efficacy. Also, it will explore the variability of instructional skills, including voice modulation, non-verbal cures, media variations, and interactive patterns, optimizing teaching delivery. It introduces the diverse range of instructional aids, and practical applications, from traditional tools to digital resource. It also helps to cultivate conducive classroom environments, managing factors affecting them, and grasp the importance of debriefing for reflective teaching practices. This course will develop comprehensive assessment tools encompassing self-assessment criteria for teachers and learners.

Major Learning Outcomes

- Recognize the importance of instructional framework
- Describe and apply the theories of transfer of learning
- Develop and apply session plan and log book
- Familiar with the variability in instructional skills
- Understand the concept of classroom management and debriefing
- Describe and demonstrate the effective use of instructional aids
- Develop and apply assessment tools

Content with Specific objectives

Specific Objectives	Contents
➢ To define and describe the instructional framework ➢ To discuss and interpret the components of instructional framework ➢ To differentiate components of instructional framework	**1. Instructional Framework 3 hours** 1.1 Concept of Instructional Framework 1.2 Components of Instructional Framework • *Instructional Models* • *Instructional Strategies* • *Instructional Methods* • *Instructional Skills*
➢ To define the concept of learning and explain Bloom's taxonomy	**2. Learning and Transfer of Learning 6 hours**

➢ To define the meaning and types of transfer of learning ➢ To explain, interpret and apply theories of transfer of learning ➢ To identify and analyze different factors influencing the transfer of learning ➢ To explore the interconnection between memory, forgetting, and the transfer of learning	2.1 Concept of learning and Bloom's Taxonomy 2.2 Theories of the transfer of learning (Mental discipline, Identical elements: Generalization) 2.3 Educational Implications of the transfer of learning 2.4 Factors affecting transfer of learning 2.5 Relationship among memory, forgetting and transfer of learning
➢ To define and explain session plan and log book ➢ To create and describe session plan and log book ➢ To interpret and analyze session plan and log book	**3. Develop Instructional Plan 8 hours** 3.1 Managing a session plan and log book • *Define a session plan and log book* • *Describe the importance and components of session plan and log book* • *Interpret form and format for developing session plan and log book*
➢ To define and explain the variability in instructional skills ➢ To explore the variations in the media and materials of instructional skills ➢ To identify and describe the patterns and levels of interactions ➢ To provide examples and explanation of instructional skills	**4. Variability in Instructional Skills 4 hours** 4.1 Variations in the instructional skills • *Voice Variations* • *Focusing, Pausing, Eye contact, Gestures, Movements* 4.2 Variations in the media and materials of instruction skills • *Visual Variations* • *Aural Variations* 4.3 Variations in the patterns and levels of interactions
➢ To identify appropriate instructional aids and demonstrate their use in the field ➢ To analyze and evaluate different instructional aids ➢ To explore the role of technology aids in modern education ➢ To discuss the importance and application of instructional aids	**5. Instructional Aid 2 hours** 5.1 Introduction to instructional aids • *Textbooks* • *Whiteboard, Smart board, Slide projector* • *Posters, Flip Charts, Videos, Photographs* • *Google Meet, Zoom, Microsoft Teams, Learning Management System (LMS), Flip* 5.2 Importance and application of instructional aids
➢ To define the classroom environment and its components ➢ To analyze the factors influencing the classroom environment	**6. Classroom Management and Debriefing 3 hours** 6.1 Concept of classroom environment 6.2 Factors affecting classroom environment

➢ To explore the techniques for managing effective classroom environment ➢ To define the concept of debriefing ➢ To describe and analyze the importance of debriefing	6.3 Managing effective classroom environment 6.4 Concept of debriefing and its importance
➢ To define self-assessment ➢ To develop and analyze the assessment tools for self-assessment ➢ To define assessment for learning and explore the types of assessment ➢ To differentiate between formative assessment and summative assessment	**7. Development of Assessment Tools** **4 hours** 7.1 Teachers' Self-Assessment • *Motivational, presentation, communication, and Questioning skills* • *Skills for small group and individual instruction* • *Developing and evaluating pupil thinking and progress* • *Classroom management and discipline* 7.2 Assessment for Learning • *Formative Assessment* • *Summative Assessment*

Major teaching and learning strategy

- Lecture classes and demonstration, Project Based Learning (PBL), and Group Work.
- Regular assessments (Five numbers)

Practical:

Field visit is mandatory for all, and course Facilitator will discuss in class when and where to go and perform the following the task in the field:

- Students will visit technical institutes and observe the teacher's teaching practice (Use of Clinical Supervision)
- Reflect the session and prepare a session plan and log book. Based on the session plan, students will carry out the session using various instructional skills and aids.
- Students will manage the classroom environment and conduct the assessment (Self-assessment and learners) of the session.
- At the end of field visit, students will prepare the field report. This report should include case description, the problem, why the problem is there, lesson plan, log book, used various instructional skills and aids, lesson learned, and conclusion

Assessment plan

In semester: 50

- First internal
- Second internal
- Project Based Learning – Practical Demonstration
- Field Visit Report

End Semester: 50

- MCQ: 10
- Subjective: 40

Evaluation Scheme for End Semester Examination

Learning Units	Hours	Marks Distribution (50 Total Marks) *		
		MCQs (1 Mark)	Short Questions (4 Marks)	Long Questions (8 Marks) *
Instructional Framework	3	1	1	1
Learning and Transfer of Learning	6	2	1	1
Develop Instructional Plan	8	3	1	1
Variability in Instructional Skills	4	1	1	1
Instructional Aids	2	1	1	1
Classroom Management and Debriefing	3	1	1	1
Development Assessment Tools	4	1	1	-
	30 hrs.	**10 x 1 = 10**	**6 x 4 = 24 * Any six**	**2 x 8 =16 * Any Two**

** There may be minor deviation in marks distribution.*

For long questions, the above module can be divided into sub-questions, if necessary.

Reference Materials:

Weber, E. (2014). *Turning learning into action*. Kogan Page Limited.

Instructional Approaches: A Framework for Professional Practice (1991). Saskatchewan Education.

Haskell, R. E. (n.d.). *Transfer of learning: Cognition, instruction and reasoning*. Academic Press.

Hill, L. (2002). Professional development strand (*Pre-instructional Skills*). Primary and Secondary Teacher Education Project.

Hill, L. (2002). Professional development strand (*Instructional Skills*). Primary and Secondary Teacher Education Project.

https://cambridge-community.org.uk/professional-development/gswafl/index.html

Occupational/Specialization Course

ITEX 215 Computer Graphics and Multimedia

Course: Computer Graphics and Multimedia	**Code:** ITEX 215
Credit Hours: 3 (2 TH + 1 PR)	**Nature:** Theory and Practical

Course Description and Goals

The student will impart the basic ideas of computer graphics and multimedia upon completion of this course. They study the basic ideas behind graphic design methods and algorithms. Additionally, they will explore various Input-Output technologies and multimedia concepts. By utilizing various graphic and multimedia tools, this course also aids students in developing their creativity.

Learning Outcomes

On completion of this course, the students will be able to:

- Explore and discover the fundamental principles of computer graphics.
- Applying 2D and 3D transformation techniques.
- Apply various clipping algorithms for hidden surface removal.
- Complement implementing graphics algorithms for visible surface determination.
- Apply and integrate multimedia elements within computer graphics.

Content with Specific Objectives

Specific objectives	Contents
• To describe the importance of Computer Graphics • To differentiate Hardware and Software Components • To Comprehend Raster Graphics and Display Systems • To discuss Scan Conversion Algorithms • To discuss Area Filling and Clipping Techniques • To apply the concepts in Computer Graphics	**Unit 1: Introduction to Computer Graphics (7 Hours)** 1. Overview and Advantages of Computer Graphics (1 hour) • Introduction to computer graphics • Advantages and importance in various fields and industries • Applications and significance in today's world 2. Hardware and Software for Computer Graphics (2 hours) • Understanding the components: display devices, input devices, graphics cards

	• Overview of display technologies (CRT, LCD, LED, etc.) • Functionality of hardware components: video controllers, random scan display systems, processors 3. Raster Graphics and Display Systems (1 hour) • Introduction to raster graphics and its role in representing images • Random Scan Display System: explanation of its working principles • Random Scan Display Processor: its functions and significance 4. Scan Conversion Algorithms (1.5 hours) • Line drawing algorithms: DDA and Bresenham's Algorithm • Circle and ellipse drawing algorithms: methods for generating circles and ellipses on a raster display 5. Area Filling and Clipping (1.5 hours) • Area-filling techniques for shapes like rectangles and ellipses • Clipping algorithms for lines, circles, and ellipses • Polygon clipping algorithms: overview and basic principles
• To Elaborate 2D Transformation Operations • To elaborate on 3D Transformation and Implement 3D Transformation Operations • To summarize the Mathematical Understanding and Problem-solving, and performing	**Unit Two: Two-Dimensional and Three-dimensional Transformation (7 Hours)** 1. 2-Dimensional Transformations (4 hours) • 2-D Translation, Rotation, Scaling (1.5 hours): Explanation of translation, rotation, and scaling operations in a 2D space.

transformations through mathematical equations and matrices • To apply transformation operations to solve problems involving spatial manipulation and orientation in both two and three-dimensional spaces • To visualize and interpret 2-D and 3-D objects	Formulas and algorithms for performing 2D translation, rotation, and scaling. • Homogeneous Coordinates (0.5 hours): Introduction to homogeneous coordinates in 2D transformations. Understanding the advantages and application of homogeneous coordinates in transformations. • Reflection and Shear Transform (1 hour): Explaining reflection transformation and its application in mirroring objects in a 2D space. Understandings shear transformation and its effects on geometric shapes in 2D. 2. 3-Dimensional Transformations (3 hours) • 3-D Translation, Rotation, Scaling (1.5 hours): Introduction to 3D transformations, covering translation, rotation, and scaling in three dimensions. Algorithms and mathematical representations for performing 3D translation, rotation, and scaling operations. • Reflection and Shear in 3D (1 hour): Application of reflection transformation in three-dimensional space. Understandings shear transformation and its effects on 3D objects.
• To explain the transformation from window to viewport coordinates, understanding coordinate system mapping principles.	**Unit Three: Clipping [7 Hours]** 1. Window to Viewport Transformation (1 hour) • Explanation of window and viewport in computer graphics

• To implement various line clipping algorithms (like Cohen-Sutherland) and apply them effectively. • To model polygon clipping algorithms (such as Sutherland-Hodgeman), applying them for complex polygon clipping. • To develop practical application skills, utilizing clipping techniques to optimize graphical output and enhance rendering efficiency. • Apply clipping algorithms to solve visibility-related problems in computer graphics, enhancing problem-solving abilities.	• Understanding the concept of transformation from the window space to the viewport space • Application of transformation matrices in window-to-viewport mapping 2. Line Clipping (3 hours) • Introduction to Clipping (0.5 hours): Understanding the purpose and significance of clipping in computer graphics. • Cohen-Sutherland Line Clipping Algorithm (1.5 hours): Explanation of Cohen-Sutherland algorithm for line clipping. Understanding the concept of region codes and the application of the Algorithm. • Practical Implementation and Examples (1 hour): Illustrating the Cohen-Sutherland algorithm with practical examples and exercises 3. Polygon Clipping (3 hours) • Introduction to Polygon Clipping (0.5 hours): Defining polygon clipping and its importance in computer graphics. • Sutherland-Hodgeman Polygon Clipping Algorithm (2.5 hours): Explanation of Sutherland-Hodgeman algorithm for polygon clipping. Step-by-step procedure and implementation of the Algorithm. • Application and Practical Demonstration (1 hour): Demonstrating the Sutherland-Hodgeman algorithm through examples and exercises.

• To differentiate the distinction between image spaces and object space techniques • To differentiate between hidden and visible surfaces in 3D scenes and models. • To explain the significance of depth comparison algorithms in identifying hidden surfaces. • To master the implementation and operation of the Z-Buffer Algorithm for hidden surface removal. • To comprehend the concept of back-face removal and its application in rendering visible surfaces. • To explore the Painter's Algorithm and its use in determining visible surfaces efficiently. • To implement the Scan-Line Algorithm and its role in hidden surface removal for complex scenes. • To explore the concepts of light, color, and various color models (RGB, CMY, YIQ) • To apply color models effectively in the rendering and representation of graphical elements.	**Unit Four: Visible Surface Determination and Computer Graphics Algorithms [14 Hours]** 1. Image Space and Object Space Techniques (1.5 hours) • Explanation of image space and object space techniques in computer graphics • Differentiating between image space and object space operations 2. Hidden Surface Removal - Depth Comparison (2 hours) • Understanding the need for hidden surface removal in computer graphics. • Depth comparison algorithms and their role in identifying visible surfaces. 3. Z-Buffer Algorithm (3 hours) • Detailed explanation of the Z-Buffer Algorithm for hidden surface removal • Implementing Z-Buffer Algorithm, understanding its advantages and limitations 4. Back-Face Removal (1.5 hours) • Introduction to back-face removal techniques. • Understanding the concept of identifying and removing hidden back-facing polygons. 5. Painter's Algorithm (2 hours) • Explanation of the Painter's Algorithm for visible surface determination. • Implementing the Painter's Algorithm, understanding its efficiency and limitations. 6. Scan-Line Algorithm (2 hours)

	• Understanding the Scan-Line Algorithm for hidden surface removal. • Practical implementation and application of the Scan-Line Algorithm in computer graphics. 7. Light, Color, and Color Models (2 hours) • Introduction to light and color in computer graphics. • Explanation of different color models (RGB, CMY, YIQ) and their applications. • Understanding the representation of color in computer graphics and its impact on rendering.
• To define multimedia in computer graphics and its various elements like text, images, audio, video, and animation. • To Explore digital media formats, compression techniques, and their significance in multimedia. • To apply interactive design principles to create engaging multimedia experiences. • To integrate text, images, audio, and video effectively for cohesive multimedia presentations. • To utilize animation techniques to enhance multimedia content. • To explore the integration of virtual reality in multimedia projects and its potential applications.	**Unit 5: Multimedia in Computer Graphics [10 Hours]** 1. Introduction to Multimedia (1 hour) • Definition and scope of multimedia in computer graphics. • Understanding multimedia elements: text, images, audio, video, and animation. • Applications and significance of multimedia in various fields. 2. Basics of Digital Media Formats (1.5 hours) • Overview of digital media formats (JPEG, PNG, GIF for images; MP3, WAV for audio; MP4, AVI for video). • Compression techniques and their importance in multimedia. 3. Principles of Interactive Multimedia Design (1.5 hours) • Understanding interactive multimedia design principles.

	• User interface considerations and user experience (UX) in multimedia applications. • Elements of effective interactive design in computer graphics. 4. Multimedia Integration Techniques (2 hours) • Integration of multimedia elements into computer graphics projects. • Techniques for combining text, images, audio, and video elements. • Creating interactive multimedia presentations using multimedia integration. 5. Animation and Multimedia (2 hours) • Exploring the role of animation in multimedia projects. • Incorporating animated elements (2D/3D) into multimedia presentations. • Techniques for synchronization of animation with other multimedia components. 6. Introduction to Virtual Reality in Multimedia (2 hours) • Defining virtual reality and its integration within Multimedia. • Understanding the applications and tools used for creating immersive VR experiences in multimedia projects. • Practical considerations and challenges in incorporating VR into multimedia.

Major Teaching and Learning Strategy

There are two groups of teaching strategies used in this course. The first group comprises general teaching strategies that can be used with most units. The second group comprises specialized teaching methods that can be used with particular units. Students will receive reading materials for each course, and all chapters use lectures, group discussions, multimedia projector use, and brainstorming for general teaching

strategy. On the other hand, a specialized teaching strategy incorporates demonstrations as a key teaching strategy during the teaching and learning process. The specific teaching method for this course will be demonstration with practical works.

Laboratory

Recommend using OpenGL library to implement graphical algorithms. The instructor has the flexibility to incorporate the laboratory topics.
The following practicals are recommended to conduct:

Implement CG algorithm using a programming language (Use OpenGL or any other programming language)

- Write a program to draw a line using the DDA algorithm.
- Write a program to draw a line using Bresenham‘s Algorithm.
- Write a program to draw a circle using the midpoint algorithm.
- Write a program to draw a circle using Bresenham‘s Algorithm.
- Write a program to draw a rectangle using the line drawing algorithm.
- Write a program to perform the 2D transformation on a line.
- Write a program to perform a shear transformation on a rectangle.
- Write a program to rotate a circle (alternatively inside and outside) around the circumference of another circle.
- Write a program to draw a car using the build graphics function and translate it from the bottom left corner to the bottom right corner of the screen.
- Write a program to draw balloons using in build graphics function and translate it from the bottom left corner to the right top corner of the screen.
- Write a program to draw a cube using in build library function and perform 3D transformations i) Translations in x, y, z directions ii) Rotation by angle 450 about the z-axis; rotation by 600 about the y-axis in succession. iii) Scaling in the x-direction by a factor of 2, scaling in the y-direction by a factor of 3.
- Write a program to implement line clipping (the Cohen-Sutherland algorithm).
- Write a program for making a Bezier curve.
- Write a program to show an animation of a ball moving in a helical path.
- Write a program to show the animation of the solar system.

Reference Material for Laboratory

https://shankarrajagopal.github.io/DOWNLOAD/CG_LAB_ALL.pdf

Assessment Plan

- **In-semester (50 marks)**
  - Internal – I
  - Internal – II
  - Assignment
  - Practical Exam/ Viva
  - Presentation
- **End-semester (50 marks)**
  - **MCQ – 10**
  - **Subjective – 40**

Marking Scheme for Final Examination:

- Section A (MCQ): 10 Questions × 1 marks
- Section B (**Short Answer Question-SAQ**) – Total 7 Questions × 4 marks (Attempt any Six)
- Section C (**Long Answer Question-LAQ**) – Total 3 Questions × 8 marks (Attempt any Two)

Learning Units	MCQ	SAQ	LAQ
Unit 1: Introduction to Computer Graphics	2	2	×
Unit Two: Two-Dimensional and Three-dimensional Transformation	2	1	1
Unit Three: Clipping	2	1	1
Unit Four: Visible Surface Determination and Computer Graphics Algorithms	2	1	1
Unit 5: Multimedia in Computer Graphics	2	2	×
	10 MCQ Questions	**Seven Questions**	**Three Questions**

References Materials

- Hearn, D., & Pauline, B. M. (2007). *Computer graphics.* Prentice Hall.
- Foley, J. D., Dam, A. V., Feiner, S. K., Hughes, J. F., Sklar, D. F., Feiner, S. K., & Skeley, K. (2014). *Computer graphics principle and practices.* Addison-Wesley Longman.
- Hearn, D., Baker, M. P. (1997). *Computer graphics* (2nd ed.). Prentice Hall of India Private Limited.
- Li, Z., Drew, M. S., Liu, J. (2014). *Fundamentals of multimedia* (2nd ed.). Springer Cham.
- Graham S., Richard S. W. Jr., & Nicholas, H., (2014) *OpenGL superbible: Comprehensive tutorial and reference* (6th ed.). Addison-Wesley Professional.
- Shreiner, D., Sellers, G., Kessenich, J. M., & Licea-Kane, B. M. (2013). *OpenGL programming guide: The official guide to learning OpenGL* (Version 4.5). Addison-Wesley Professional.
- Marschner, S., Shirley, P. (2016). *Fundamentals of computer graphics.* CRC Press.
- Patric C. (2015). *WebGL insights.* CRC Press.

ITEX 203 System Analysis and Design

Course: System Analysis and Design	**Course Code:** ITEX 203
Credit Hours: 3 (2 TH + 1 PR)	**Nature:** Theory and Practical

Course Description

System analysis and design relates to shaping organizations, improving performance, and achieving objectives for profitability and growth. The emphasis is on systems in action, the relationships among subsystems, and their contribution to meeting a common goal. This course mainly focuses on different aspects of system analysis and design, such as *foundation, planning, analysis, design, implementation, and maintenance*.

Learning outcome

On completion of this course, the students will be able to:

- Identify the elements of the System Development Life Cycle (SDLC).
- Identify the major tasks of each phase of the SDLC.
- Explain the process of managing an information systems project.
- Develop a solid foundation in designing a system, as well as the practical skills necessary to implement and maintain the systems.
- Relate several subsystems to integrate and develop a single system.
- Apply concepts, skills, and techniques required to analyze and design the system.

Specific Objectives	Contents
• To define and describe information systems and their types. • To develop a Systems Development Life Cycle (SDLC). • To explore the Systems Development Process. • To discuss Traditional Waterfall SDLC. • To define and describe different approaches to improving development, such as *Rapid Application Development (RAD), Computer Aided Software Engineering (CASE) tools, Service-Oriented Architecture (SOA), Agile Methodologies, eXtreme Programming, Object Oriented Analysis and Design*.	**Unit 1: The Systems Development Environment (10 hours)** Information System and its Types, Systems Development Life Cycle (SDLC), Systems Development Process, Traditional Waterfall SDLC, Approaches for Improving Development (CASE Tools, Rapid Application Development, Service-Oriented Architecture, Agile Methodologies, eXtreme Programming, Object-Oriented Analysis and Design)

• To define and describe the project identification and selection process of systems development projects. • To explain the relationship between corporate strategic planning and information systems planning. • To describe the use of information systems planning to assist in identifying and selecting systems development projects. • To discuss steps involved in the project initiation and planning process. • To explore various methods for assessing project feasibility. • To develop and review the Baseline Project Plan.	**Unit 2: Planning (8 hours)** System Development Projects: *Identification and Selection:* Introduction, Systems Development Projects, Corporate and Information Systems Planning *Initiation and Planning*: Introduction, Initiating and Planning Systems, Development Projects, Initiating and Planning Information System (IS) process, Development Projects, Project Feasibility, Baseline Project Plan
• To discuss performing requirements determination. • To define and describe the traditional, contemporary, and radical methods for determining system requirements. • To explore requirement management tools. • To discuss determining requirements using agile methodologies. • To describe the logical modeling of processes by studying examples of data flow diagrams. • To analyze and interpret decision tables to represent the logic of choice in conditional statements. • To define and explore the techniques of gathering information for conceptual data	**Unit 3: Analysis (10 hours)** *System Requirements:* Introduction, Requirements Determination, Traditional Methods for Determining Requirements, Contemporary Methods for Determining System Requirements, Radical Methods for Determining System Requirements, Requirement Management Tools, Requirements Determination Using Agile Methodologies *System Process Requirements:* Introduction, Process Modeling, Data Flow Diagramming Mechanics, Data Flow Diagramming in the Analysis Process, Modeling Logic with Decision Tables

modeling in the overall analysis and design of an information system. • To develop and interpret an Entity-Relationship (E-R) diagram representing supertypes, subtypes, and business rules to represent common business situations. • To discuss the role of packaged conceptual data models in database systems.	*System Data Requirements:* Introduction, Conceptual Data Modeling, Gathering Information for Conceptual Data Modeling, Introduction to E-R Modeling, Conceptual Data Modeling and the E-R Model, Representing Supertypes and Subtypes, Business Rules, Role of Packaged Conceptual Data Models - Database Patterns
• To define and describe database design using a relational database model. • To discuss different normalization techniques in database design. • To discuss transforming an entity-relationship (E-R) diagram into an equivalent set of well-structured (normalized) relations. • To explore merging relations techniques in database design. • To discuss the relationship of physical files in database design. • To define and explore designing fields and physical tables in database design. • To explain the process of designing forms and reports and formatting them. • To describe and interpret the usability of forms and reports. • To discuss the process of designing interfaces and dialogues and the deliverables for their creation. • To explore interaction methods and devices to develop interfaces and dialogs.	**Unit 4: Design (11 hours)** *Designing Databases:* Introduction, Database Design, Relational Database Model, Normalization, Transforming E-R Diagrams into Relations, Merging Relations, Physical File and Database Design, Designing Fields, Designing Physical Tables *Designing Forms and Reports:* Introduction, Designing Forms and Reports, Formatting Forms and Reports, Usability *Designing Interfaces and Dialogues:* Introduction, Designing Interfaces and Dialogues, Interaction Methods and Devices, Designing Interfaces and Dialogues in Graphical Environments

• To develop graphical user interfaces using interfaces and dialogs.	
• To define and describe system implementation. • To discuss the role of software application testing in system implementation. • To explore the installation process and document the system. • To discuss developing the training manual to support the users. • To describe the organizational issues in system implementation. • To discuss the importance of maintaining the information system. • To describe the techniques to conduct system maintenance.	**Unit 5: Implementation and Maintenance (6 hours)** *System Implementation:* Introduction, System Implementation, Software Application Testing, Installation, Documenting the System, Training and Supporting Users, and Organizational Issues in Systems Implementation. *System Maintenance:* Introduction, Information Systems Maintenance

Major Teaching and Learning Strategy

Lecture classes: Instructors will provide information of the contents with several real examples to the students throughout the regular lectures.

Demonstration: The instructor will demonstrate several examples to the students to make them clear on their understanding the concepts.

Regular assessments: Students will be provided with several assignments related to different units.

Presentation: Students will be provided with several topics to present. This will basically share the information to identify the student's understanding.

Problem-based assessment: A group will be formed to identify the problem and implement the suitable concepts discussed in the classroom to design the system.

Case study: Both instructors and students will be actively engaged. They will be presenting several cases in regular lectures. Thus will be inquiry-based and also collaborative.

Project and Practical

Students need to form a group and conduct a mini-project. This includes identifying a real problem and implementing several concepts discussed in the classroom to design the system. This is a group work. Hence, a collaborative approach will be encouraged. Some examples include conducting a case study of different systems such as Library Management Systems, Inventory Management Systems, Student Profile Systems, Seat Reservation Systems, eSewa, Khalti, etc.

Assessment Plan

In semester: 50

Written exam, presentation, demonstration, project report, assignments, objectives, attendance, viva.

End Semester: 50

Objectives, written exam

Marking Scheme for Final Examination

Learning Units	Multiple Choice Items (1 Mark each)	Short Answer Questions (4 Marks each)	Long Answer Questions (8 Marks each)
Unit 1: The Systems Development Environment	2	1	1
Unit 2: Planning	2	2	
Unit 3: Analysis	2	1	1
Unit 4: Design	2	1	1
Unit 5: Implementation and Maintenance	2	2	
Total	10	7	3

References Book and Materials

Hoffer, J. A., George, J. F., Valacich, J. S., & Valacich, J. A. (2017). *Modern systems analysis and design* (8th ed.). Pearson Education.

Kendall, K. E., & Kendall, J. E. (2020). *System analysis and design* (10th ed.). Pearson.

Tilley, S. (2020). *System analysis and design* (12th ed.). Cengage Learning, Inc.

Tilley, S., & Rosenblatt, H. (2017). *System analysis and design* (11th ed.). Cengage Learning.

Whitten, L. J., & Bently, L. D. (2005). *Systems analysis and design methods* (7th ed.). McGraw-Hill.

ITEX 220 Operating System

Course Title: Operating System	**Code:** ITEX 220
Credit Hours: 3 (2 TH + 1 PR)	**Nature:** Theory +Practical

Course Description and Goals

This course encompasses both fundamental design principles and the implementation of operating systems. It emphasizes providing programs with abstracted access to hardware resources. Topics covered include file systems, virtual memory, memory management, process scheduling, and a case study involving the Linux operating system.

Learning Outcomes

On completion of this course, the students will be able to:

- Understand the fundamental components of an operating system.
- Comprehend how an operating system virtualizes CPU and memory.
- Discuss various process scheduling and swapping strategies.
- Understand the concept of Virtual Memory.
- Familiarize them with Device and I/O management.
- Comprehend the organization of file structures in Linux-based operating systems.

Content with Specific Objectives

Specific objectives	Contents
• To define the operating system and illustrate its pivotal role in managing hardware resources and facilitating user applications. • To describe the diverse perspectives of the OS: application view for developers/users and the system view focused on resource management and service provision. • To explore the implementation view, detailing the interaction intricacies between the OS and hardware components. • To provide an overview of the historical evolution of operating	**Unit 1: Introduction (5 Hours)** 1. Definition of Operating System: Explanation of what an operating system is. 2. Views of Operating System: Application View: How the OS appears to application developers/users. System View: How the OS manages resources and provides services. Implementation View: How the OS is implemented and interacts with hardware. 3. History and Evolution of Operating System: Overview of the historical development and Evolution of operating systems. Milestones and major advancements in OS design and functionalities.

systems, highlighting significant milestones and key developmental stages. • To discuss various types of operating systems based on user count and task handling, encompassing single-user vs. multi-user and single-tasking vs. multi-tasking systems. • To introduce specialized OS types such as real-time systems, network operating systems, and embedded systems, outlining their distinct characteristics and functionalities. • To explain the significance of system calls in enabling communication between applications and the OS, detailing their execution and management processes.	4. Types of Operating Systems: Classification and categorization of OS types: Single-user vs. multi-user systems. Single-tasking vs. multi-tasking systems. Real-time systems, network operating systems, embedded systems, etc. 5. System Call and Call Handling: Explanation of system calls and their importance. How system calls are handled by the operating system. Examples of common system calls and their functionalities. 6. System Programs and OS Structures: Overview of system programs and their roles in an operating system. Discussion of different OS structures (monolithic, microkernel, etc.). The role and functionalities of the shell in interacting with the OS. 7. Virtual Machine: Introduction to virtualization and its importance in modern computing. Explanation of virtual machines and their role in OS environments. Benefits and applications of using virtual machines.
• To differentiate between processes and programs, highlighting their distinct functionalities and roles within an operating system. • To describe the process model, outline various states (new, ready, running, waiting, and terminated),	**Unit Two: Process, Process Management and Scheduling (11 hrs.)** 1. Processes and Threads: • Understanding the distinction between a process and a program. • Introduction to multiprogramming and its significance.

and emphasize the significance of the Process Control Block (PCB). • To explore threads, highlighting differences between threads and processes, including discussions on user and kernel space threads. • To discuss Inter-Process Communication (IPC) mechanisms and their crucial role in enabling communication between processes. • To explain fundamental concepts like race conditions, critical sections, and mutual exclusion in concurrent programming. • To explore techniques for achieving mutual exclusion, including Busy Waiting, Sleep and Wakeup mechanisms, Semaphores, Monitors, and Message Passing. • To detail scheduling objectives and methodologies, covering batch scheduling algorithms (FCFS, SJF, SRTN), Round-Robin scheduling, priority-based scheduling, and the concept of multiple queues for scheduling diverse types of processes.	• Process model, states (such as new, ready, running, waiting, terminated), and Process Control Block (PCB). • Exploring threads, differences between threads and processes, user and kernel space threads. • Inter-process communication (IPC) and its mechanisms. • Concepts of race conditions, critical sections, and mutual exclusion. • Various techniques for achieving mutual exclusion: a) Busy Waiting (Disabling Interrupts, Lock Variables, Strict Alteration, Peterson's Solution, Test and Set Lock).Sleep and Wakeup mechanisms. Semaphores, Monitors, and Message Passing. b) Classical IPC problems: Producer-Consumer, Sleeping Barber, Dining Philosopher Problem. 2. Process Scheduling: • Goals of process scheduling. • Batch Scheduling algorithms: a) First-Come, First-Served (FCFS). b) Shortest Job First (SJF). c) Shortest Remaining Time Next (SRTN). 3. Interactive Scheduling: • Round-Robin scheduling. • Priority-based scheduling. • Multiple Queues for scheduling.
• To define deadlock and its significance in operating systems.	**Unit Three: Deadlock and Deadlock Handling (6 hrs.)** 1. Introduction to Deadlock:

• To characterize deadlock by identifying necessary conditions and understanding preemptable vs. non-preemptable resources. • To explain the Resource-Allocation Graph (RAG) and its role in visualizing resource allocation and potential deadlocks. • To discuss strategies for deadlock handling, including the Ostrich Algorithm. • To present deadlock prevention techniques and methods to ensure at least one condition for deadlock cannot hold. • To describe deadlock avoidance strategies to prevent the system from encountering deadlocks. • To Explain deadlock detection algorithms and recovery mechanisms once a deadlock is detected.	• Definition and significance of deadlock in an operating system. • Deadlock characterization: identifying necessary conditions for a deadlock to occur. • Understanding preemptable and non-preemptable resources. • Resource-Allocation Graph (RAG) and its use in visualizing resource allocation and potential deadlocks. • Conditions required for deadlock occurrence in a system. 2. Deadlock Detection and Handling: • Strategies for deadlock handling: a) Introduction to the Ostrich Algorithm. b) Deadlock prevention techniques: methods to ensure that at least one of the necessary conditions for deadlock cannot hold. c) Deadlock avoidance: dynamic strategies to prevent the system from entering an unsafe state and encountering deadlocks. d) Deadlock detection for multiple resources and recovery mechanisms: Algorithms to detect deadlocks in a system. **e)** Recovery strategies once a deadlock is detected (e.g., process termination, resource preemption, etc.).
• To provide an overview of memory management in operating systems.	**Unit Four: Memory Management (6 hrs.)** 1. Introduction to Memory Management • Overview of memory management in operating systems.

• To differentiate between logical and physical addresses, explaining their roles in memory access. • To explain the concept of swapping and its significance in moving processes between main memory and secondary storage. • To discuss contiguous memory allocation techniques, explaining their mechanisms and limitations. • To analyze the advantages and limitations of contiguous memory allocation strategies. • To introduce paging as a memory management technique, detailing the use of page tables, page faults, and page replacement algorithms. • To explain segmentation as an alternative memory management scheme, outlining its benefits and challenges.	• Understanding the difference between logical and physical addresses. • Concepts of swapping: moving processes between main memory and secondary storage. 2. Memory Allocation Techniques: • Contiguous Allocation: Explanation of contiguous memory allocation techniques. Discussing advantages and limitations of contiguous allocation. 3. Non-Contiguous Memory Allocation: • Paging: Introduction to paging as a memory management technique. Understanding page tables, page faults, and page replacement algorithms. • Segmentation: Overview of segmentation as another memory management scheme. Benefits and challenges associated with segmentation.
• To provide an overview of virtual memory, emphasizing fundamental concepts and its role in operating systems. • To explain the functioning of virtual memory within an OS and highlight the significance and benefits of its implementation. • To detail the operational mechanism of virtual memory management, including its	**Unit Five: Virtual Memory (6 hrs.)** 1. Introduction to Virtual Memory: • Overview and fundamental concepts of virtual memory. • Explanation of how virtual memory functions within an operating system. • Significance and benefits of using virtual memory. 2. Working Mechanism of Virtual Memory: • Detailing the operational mechanism of virtual memory management.

interaction with physical memory and secondary storage. • To analyze the benefits and drawbacks of implementing virtual memory, emphasizing its impact on system performance and efficiency. • To explain demand paging within the context of virtual memory and how it operates in managing memory resources. • To Assess the performance implications of demand paging in virtual memory systems. • To revisit various page replacement algorithms (FIFO, Second Chance, Optimal, LRU, LFU) concerning their relevance and application within virtual memory management. • To introduce frame allocation strategies specifically within the context of virtual memory, categorizing different methods used in virtual memory management. • To define thrashing in the context of virtual memory systems, identify its causes and effects, and propose remedies to mitigate its impact.	• How it interacts with physical memory and secondary storage. 3. Advantages and Disadvantages of Virtual Memory: • Understanding the benefits and drawbacks of implementing virtual memory. • Discussing the impact on system performance and efficiency. 4. Demand Paging in Virtual Memory: • Reiteration of demand paging within the context of virtual memory. • How demand paging operates in the virtual memory environment. • Assessing performance implications of demand paging in virtual memory systems. 5. Page Replacement Algorithms in Virtual Memory: • Revisiting page replacement algorithms in the context of virtual memory. • FIFO, Second Chance, Optimal, LRU, LFU algorithms and their relevance to virtual memory management. 6. Frame Allocation Categories in Virtual Memory: • Introduction to frame allocation strategies within virtual memory. • Categorization of frame allocation methods, specifically within the context of virtual memory management. 7. Thrashing in Virtual Memory Systems: • Understanding thrashing specifically in the context of virtual memory systems. • Causes, effects, and remedies for thrashing in a virtual memory environment.

• To describe the fundamental concepts of a file system, including its structure and organization. • To explain access mechanisms for reading and writing files within the file system. • To explore different directory structures (e.g., hierarchical, flat) and their implementation within file systems. • To discuss file protection mechanisms, including access permissions, security measures, and protection mechanisms. • To ensure data consistency and define file operation semantics within the file system. • To provide an overview of device management in operating systems, emphasizing its importance in system functionality. • To explain I/O (Input/Output) interfaces, detailing how devices are interfaced with the system for I/O operations. • To explore the role and functionalities of the kernel in managing I/O operations within the Kernel I/O Subsystem.	**Unit Six: File and Device Management (7 hrs.)** 1. File System Concepts: • Understanding the fundamental concepts of a file system. • Access mechanisms: How files are accessed, read, and written within the file system. • Directory structure: Exploring different types of directory structures (e.g., hierarchical, flat) and their organization within file systems. • Protection mechanisms: Discussing file access permissions, security, and protection mechanisms. • Consistency and Semantics: Ensuring data consistency and defining file operation semantics within the file system. 2. Introduction to Device Management: • Overview of device management in operating systems. • Understanding I/O (Input/Output) interfaces: How devices are interfaced with the system for I/O operations. • Kernel I/O Subsystem: Exploring the role and functionalities of the kernel in managing I/O operations.

• To explore the historical background and Evolution of UNIX and Linux, highlighting key milestones and influential developments. • To explore the concept of kernel modules in Unix-like systems, explaining their roles and how they extend the kernel's functionalities. • To investigate how Linux and UNIX systems handle processes, covering their creation, scheduling, and termination mechanisms. • To examine the process scheduling algorithms and strategies implemented in Linux and UNIX systems, detailing their functionalities. • To describe the mechanisms for communication between processes within Linux and UNIX environments, exploring various IPC mechanisms. • To explore memory management techniques, including virtual memory and page handling, implemented in Linux and UNIX systems. • To analyze how file systems are structured and managed in Linux and UNIX environments, including device management and I/O operations.	**Unit Seven: OS Case Study (4 hrs.)** 1. Linux and Unix: • History: Understanding the historical background and Evolution of UNIX and its derivative, Linux. • Kernel Modules: Exploring the concept of kernel modules in Unix-like systems, their roles, and how they extend the kernel's functionalities. • Process Management: Investigating how Linux and UNIX handle processes, their creation, scheduling, and termination. • Scheduling: Examining the process scheduling algorithms and strategies implemented in Linux and UNIX systems. • IPC (Inter-Process Communication): Understanding the mechanisms for communication between processes within Linux and UNIX environments. • Memory Management: Exploring memory management techniques, including virtual memory and page handling in Linux and UNIX systems. • File and Device Management: Analyzing how file systems are structured and managed in Linux and UNIX, including device management and I/O operations.

Major Teaching and Learning Strategy

There are two groups of teaching strategies used in this course. The first group comprises general teaching strategies that can be used with most units. The second group comprises specialized teaching methods that can be used with particular units. Students will receive reading materials for each course, and all chapters use lectures, group discussions, multimedia projector use, and brainstorming for general teaching strategy. On the other hand, specialized teaching strategies incorporate demonstrations as a key teaching strategy during the teaching and learning process. The specific teaching method for this course will be a demonstration of practical works.

Practical Works (But not limited to)

- Experiments to understand the operating system (Linux) installation process, file system portioning, and dual boot setup.
- Experiment with learning the command line interface (shell) and exploring various UNIX commands.
- Writing programs to create and execute shell scripts.
- Program to implement various algorithms for process scheduling.
- Program to implement various algorithms for page replacement activity in memory management.
- Writing programs to use inter-process communication constructs (File sockets/ Shared memory).

Assessment Plan

a. **In-semester (50 marks)**
   - **i.** Internal – I
   - **ii.** Internal – II
   - **iii.** Assignment
   - **iv.** Practical Exam/Viva

b. **End-semester (50 marks)**
   - i. MCQ – 10
   - ii. Subjective – 40

<u>Marking Scheme for Final Examination:</u>

- Section A (**MCQ**): 10 Questions × 1 marks
- Section B (**Short Answer Question-SAQ**) – Total 7 Questions × 4 marks (Attempt any Six)
- Section C (**Long Answer Question-LAQ**) – Total 3 Questions × 8 marks (Attempt any Two)

Learning Units	MCQ	SAQ	LAQ	Remarks
Unit 1: Introduction	1	1	×	• The final examination structure requires one Short Answer Question (SAQ) and one Long Answer Question (LAQ) specifically from Chapter 2. • Other two Long Answer Questions (At least one question from Chapters 3, 4, or 5) • An extra Short Answer Question (SAQ) will be asked, originating from either Chapter 3, 4, or 5, provided that this SAQ does not come from the same chapter for which a Long Answer Question (LAQ) has been asked.
Unit Two: Process, Process Management and Scheduling	2	1	1*	
Unit Three: Deadlock and Deadlock Handling	2	1	1*	
Unit Four: Memory Management	2	1	1*	
Unit Five: Virtual Memory	2	1	1*	
Unit Six: File and Device Management	1	1	×	
	10 MCQ Questions	**Seven Questions**	**Three Questions**	

Reference Materials

- Tanenbaum, A. S., & Bos, H. (2023). *Modern operating systems* (5th ed.). Pearson.
- Silberschatz, A., Galvin, P. B., & Gagne, G. (2018). *Operating system concepts* (10th ed.). Wiley.
- Anderson, T., & Dahlin, M. (2012). *Operating systems: Principles and practice*. Anderson & Dahlin.
- Love, R. (2012). *Linux kernel development* (3rd ed.). Addison-Wesley.

ITEX 211 E-Learning Management System

Course Title: E-Learning Management System	**Course Code:** ITEX 113
Credit Hours: 2 (1 TH + 1 PR)	**Nature:** Theory and Practical

Course Description and Goals

The course titled "ITEX 211 E-Learning Management System" addresses the increasing importance of digital education in the contemporary global context. The significance of e-learning and Learning Management Systems (LMS) has grown more apparent as conventional learning approaches encounter obstacles. The utilization of Learning Management Systems (LMS) in the context of e-learning provides a means of delivering education that is adaptable and easily available. This mode of education allows learners to interact with educational materials at their preferred speed and from various geographical places. Within the present global landscape, characterized by the widespread use of remote and online education, this course aims to tackle the pressing necessity for individuals to grasp and utilize the capabilities of Learning Management Systems (LMS) in order to facilitate the delivery of online courses. The course's current and future significance stems from its alignment with modern educational methodologies, which prioritize the crucial role of Learning Management Systems (LMS) in providing high-quality education, improving accessibility, and equipping learners for the ever-changing digital learning environment.

Note for Course Duration: 15 Hours Theory Class + 30 Hours Practical Class

The 30 Hours of the practical class comprises 15 hours of hands-on lab sessions and an additional 15 hours allocated for engaging in various external online courses, projects, resources, and practical tasks.

Learning Outcomes

On completion of this course, the students will be able to:

- Understand the essence of e-learning management systems (LMS).
- Identify the key components and functionalities of LMS.
- Explore various LMS platforms and their features.
- Understand the process of creating and managing online courses using LMS.
- Implement best practices for effective online learning design and delivery.
- Evaluate the impact of LMS on education and training.

Content with specific objectives

Specific Objectives	Learning Contents

● Explain the evolution of e-learning and recognize current trends. ● Define the concept of Learning Management Systems (LMS) and assess their benefits and challenges. ● Identify the applications of LMS in education and training and distinguish between different types of E-Learning Management Systems.	**Module 1: Introduction to E-Learning and LMS (6 hours)** ● Evolution of e-learning & emerging trends ● What are Learning Management Systems (LMS)? ● Benefits and challenges of using LMS ● Applications of LMS in education and training ● Types of E-Learning Management Systems
● Identify and explain the essential components that make up Learning Management Systems. ● Implement best practices for user management within an LMS. ● Ensure secure authentication processes for a protected learning environment. ● Organize course materials in a structured and accessible manner.	**Module 2: LMS Components, Functionalities, and Course Design (12 hours)** ● Overview of LMS components ● User management and authentication ● Content delivery and organization ● Instructional design for online learning ● Development of online course materials
● Understand and execute the steps involved in implementing an LMS. ● Demonstrate the ability to integrate an LMS seamlessly with existing educational systems. ● Customize and personalize an LMS to suit specific educational requirements. ● Adapt digital pedagogy for online delivery. ● Foster learner engagement in the online environment through interaction and collaboration.	**Module 3: LMS Selection, Implementation, and Pedagogical Considerations (12 hours)** ● Factors to consider when selecting an LMS ● LMS implementation process ● LMS integration with existing systems ● Customization and personalization of LMS ● Designing effective online courses: - Adapting digital pedagogy for online delivery - Principles of effective online learning design - Engaging learners in the online environment - Promoting collaboration and interaction

• Gain hands-on experience in navigating and using Learning Management Systems (LMS) for online education. • Develop proficient skills in crafting compelling and effective materials for online courses. • Implement a variety of assessment techniques tailored for digital learning environments. • Apply strategies for efficient course management and facilitation in the online learning environment. • Develop and implement comprehensive evaluation plans for courses conducted through an LMS.	**Module 4 (Practical): Creating and Managing Online Courses (15 hours)** • Hands-on experience with an LMS platform • Development of online course materials • Assessment techniques for online courses • Course management and facilitation strategies • Designing and implementing an evaluation plan for an LMS-based course

Unit-wise Activities Plan

Module I - Assignments/Projects

a) Evolution of E-Learning Timeline:

Assignment: Design a timeline that illustrates the key milestones in the evolution of e-learning. Highlight advancements in technology and trends that have influenced how we engage with digital education.

b) LMS Case Study Analysis:

Assignment: Select a real-world case study involving an educational institution or organization that has effectively implemented a Learning Management System (LMS). Evaluate and present the advantages, challenges, and overall impact of the LMS on the learning environment within that institution.

Key Reading Resources and Tools

- The evolution of eLearning (https://www.ispringsolutions.com/blog/the-evolution-of-e-learning-from-baby-steps-to-giant-leaps
- Learning Management System (LMS) used with online instruction (https://files.eric.ed.gov/fulltext/EJ1286531.pdf)
- Global E-Learning Market Size (https://www.linkedin.com/pulse/global-e-learning-market-size-share-projections/)

Module II - Assignments/Projects

a) LMS Feature Analysis:

Assignment: Select a popular Learning Management System (e.g., Moodle, Canvas) and conduct an in-depth analysis of its key features. Evaluate how these features contribute to effective online learning.

b) Instructional Design Showcase:

Assignment: Develop a brief instructional design plan for an online module. Include strategies for content presentation, engagement, and assessment within the context of an LMS.

c) Complete any one of the below Online Course.

- Ubiquitous Learning and Instructional Technologies (https://www.coursera.org/learn/ubiquitouslearning?)
- e-Learning Ecologies (https://www.coursera.org/learn/elearning)

Notes:

- *The above assignments and projects can be assigned as either individual or group work.*
- *Participation in activity C is mandatory for each student to gain an understanding of the structure and management of an online course. It provides valuable insights into exploring various features and components of online courses. Additionally, students are required to create a report detailing different aspects, including the organization of course materials, assessment methods, enrollment procedures, authentication systems, etc., based on the online course they have completed.*

Key Reading Resources and Tools

- Zaharias, P., & Pappas, C. (2016). Quality management of learning management systems: A user experience perspective. *Current Issues in Emerging eLearning*, *3*(1), 5.
- Learning Management System- All you need to Know (https://digitalmahbub.com/learning-management-system/)
- Moodle, Google Classroom, Coursera Course

Module III - Assignments/Projects

a) Complete this course: How to Create an Online Course: The Official Udemy Course (https://www.udemy.com/course/official-udemy-create-course/)

b) Customization and Personalization Strategies:

Assignment: Develop a plan for customizing and personalizing an LMS to meet the unique needs of a hypothetical educational setting. Include considerations for diverse learning styles and preferences.

c) Online Course Design Challenge:
Project: Design an online course using an LMS, considering digital pedagogy and principles of effective online learning design. Include strategies to enhance learner engagement and collaboration.

d) Interactive LMS Workshop:
Project: Develop and conduct a workshop demonstrating the customization features of an LMS. Showcase how educators can personalize the platform to create engaging and effective online learning experiences.

Notes:

- *Participation in activity A is mandatory for each student.*
- *The above assignments and projects can be assigned as either individual or group work. A minimum of two out of the last three suggested assignments or projects (b,c,d) should be selected for implementation.*
- *Modules 3 and Module 4 (Practical) should be taught concurrently, as Module 3 primarily covers the theoretical aspects of online learning and LMS, while Module 4 guides students through its practical implementation.*

Key Reading Resources and Tools

- How To Choose The Right LMS Platform (https://elearningindustry.com/how-to-choose-the-right-lms-platform-ultimate-guide)
- Designing Effective Online Courses (https://red.mnstate.edu/ijgll/vol2/iss1/5/)
- Moodle, Google Classroom

Module IV (Practical) - Assignments/Projects

a) LMS Platform Exploration Report:
Assignment: Research and compile a report on the features and functionalities of a specific LMS platform. Evaluate its suitability for creating and managing online courses, considering factors such as user-friendliness and customization options.

b) Online Course Materials Showcase:

Assignment: Develop a sample module for an online course. Showcase engaging materials, including text, multimedia, and interactive elements. Include a rationale for the chosen content and its alignment with online teaching principles.

c) Assessment Strategy Analysis:
Assignment: Analyze and compare different assessment strategies for online courses. Create a guide outlining the implementation of diverse assessment techniques within an LMS, considering factors like formative and summative assessments.

d) Online Course Creation Project:
Project: Individually or in groups, design and create an entire online course using an LMS. This project involves crafting engaging materials, implementing assessments, and developing strategies for course management.

Notes:

- *The above assignments and projects can be assigned as either individual or group work. A minimum of two out of the four suggested assignments or projects should be selected for implementation.*
- *Modules 3 and 4 should be taught concurrently, as Module 3 primarily covers the theoretical aspects of online learning and LMS, while Module 4 guides students through its practical implementation.*

Key Reading Resources and Tools

- Gamage, S. H., Ayres, J. R., & Behrend, M. B. (2022). A systematic review of trends in using Moodle for teaching and learning. *International Journal of STEM Education*, *9*(1), 1-24.
- Moodle, Google Classroom

Assessment Plan

- Internal Assessment: 50%
  - √ Classroom Activities: 15%
  - √ Assignments: 15%
  - √ Attendance: 10%
  - √ Presentation: 10%
- End Semester Assessment: 50%
  - Comprehensive E-Learning Implementation Project (20%)

- Open-book Exam (15%)
- Theme-based article (Min. 1000 words) and post in a classroom blog (5%)
- Viva voce (10%)

Major Teaching and Learning Strategies

- **Interactive Sessions**: Incorporate interactive sessions using tools within an LMS or external platforms to demonstrate features and engage students in hands-on activities.
- **Hands-on Workshops**: Conduct practical workshops, allowing students to interact with various LMS platforms, explore features, and gain firsthand experience in creating sample courses.
- **Assignments and Projects**: Assign tasks like creating an Evolution of E-Learning Timeline or analyzing a real-world LMS case study to enhance research and analytical skills.
- **Online Course Completion**: Require students to complete an actual online course (e.g., from Coursera) to provide insights into the learner's perspective and understand online course structures.
- **Practical Labs**: Allocate time for practical lab sessions, allowing students to experiment with LMS functionalities in a controlled environment.
- **Guest Speakers**: Invite industry experts or professionals from educational institutions to share practical insights and experiences in successfully implementing LMS.
- **Reflection and Feedback**: Include reflective activities where students assess their learning and experiences, providing constructive feedback on assignments and projects.

Reference Materials

- Adams Becker, S., Estrada, V., & Hernandez, A. (2023). Edtech for the K-12 Classroom: A Guide to Integrating Technology for Learning and Teaching, Second Edition. International Society for Technology in Education (ISTE).
- Boettcher, Judith V., and Rita-Marie Conrad. *The online teaching survival guide: Simple and practical pedagogical tips*. John Wiley & Sons, 2021.

ITEX 216 Research Methods for Computing and Technology

Course Title: Research Methods for Computing and Technology	**Course Code:** ITEX 216
Credit Hours: 2 (1 TH + 1 PR)	**Nature:** Theory and Practical

Course Description and Goals

Research Methods in Information Technology is a course designed to equip students with the knowledge and skills to conduct research in information technology and related fields. The course covers various aspects of research, such as formulating research problems, conducting literature reviews, designing research methods, collecting and analyzing data, and writing research proposals and reports. The course also addresses ethical issues and intellectual property rights in technology research.

The course consists of six units, each focusing on a different aspect of research. The first unit introduces the concepts of qualitative and quantitative research in computing research and how to formulate research questions. The second unit guides the students on how to conduct effective literature reviews in computing and technology and how to identify research gaps. The third unit covers various research methods, such as experimental, non-experimental, survey, and case study methods, and how to select appropriate samples and variables. The fourth unit teaches the students how to analyze quantitative and qualitative data using software tools such as SPSS and thematic analysis. The fifth unit discusses the ethical principles and intellectual property rights in technology research and how to ensure privacy, anonymity, confidentiality, and avoid plagiarism. The sixth unit helps the students to structure their research proposals and reports in technology studies and to use academic writing style and citation techniques.

Learning Outcomes

On completion of this course, the students will be able to:

- Demonstrate the fundamental knowledge of qualitative and quantitative research in Information Technology
- Formulate research problem, purposes, and questions
- Conduct literature review
- Demonstrate knowledge and skills in quantitative and qualitative research methods.
- Analyze and apply data collection approaches to collect the data and use appropriate methods to analyze the data.
- Apply ethical considerations to the research in information technology
- Develop and present the preliminary research report

Content with specific objectives

Specific objectives	Contents

1. Differentiate between qualitative and quantitative research approaches. 2. Conceptualize the nature of data in qualitative and quantitative research 3. Describe, exemplify, and apply scales of measurement 4. Develop or formulate a research problem, purpose, and questions	**Unit One: Introduction to Research (4 hours)** • Overview of Research Methods in Information Technology • Concept of qualitative and quantitative research in computing research • Nature of Qualitative and Quantitative Data [Scales of Measurement (Nominal, Ordinal, Interval, Ratio Scales)] • Formulating Research Problem, Research Purpose(s), and Research Question(s)
1. Conceptualize the ideas of the literature review 2. Differentiate between literature review in qualitative and quantitative research 3. Conduct literature review and identify the research gap	**Unit Two: Conducting Review of Literature (4 hours)** • Concepts of Literature Review • Literature Review in Quantitative and Qualitative Research in Information Technology • Conducting effective literature reviews in computing and technology • Identifying research gaps

1. Explore and conceptualize research methods in quantitative and qualitative research design
2. Interpret and analyze the ideas of population, sample, and sampling techniques in quantitative research
3. Interpret and apply the concepts of variables in research
4. Differentiate between experimental and non-experimental research design
5. Conceptualize the ideas of the survey research method
6. Develop data collection tools (questionnaires)
7. Interpret the use of reliability and validity in quantitative research
	Unit Three: Research Design (13 hours)
	• Quantitative Research Design
	  ○ Population and Sampling in Quantitative Research (Probability Sampling Approaches (Simple random sampling, Systematic Sampling, stratified sampling, cluster sampling)
	  ○ Variables in Quantitative Research (Dependent and Independent variables)
	  ○ Experimental and Non-Experimental Research Design
	  ○ Survey Research Method (Concepts, Data Collection Strategies, etc.)

8. Interpret and exemplify qualitative research in information technology 9. Describe the ideas of sampling in qualitative research method 10. Explain the case study and use it in qualitative research 11. Interpret the ideas of trustworthiness in qualitative research	○ Developing Data Collection Tools (Questionnaires and Experimental tools) ○ Reliability and Validity in Quantitative Research • Qualitative Research Methods ○ Sampling in Qualitative Research (Purposive and Snowball Sampling) ○ Case Study Research (Concepts, Data Collection Strategies, Planning, Applying Case Studies, Writing Case Studies) ○ Trustworthiness (Credibility, Transferability, confirmability, and dependability) in Qualitative Research
1. Use descriptive and inferential statistical methods to analyze quantitative data 2. Use theme generation process and meaning-making process to analyze qualitative data	**Unit Four: Analyzing Data (4 hours)** • Analyzing Quantitative Data using SPSS ○ Descriptive (Frequency, Central Tendency, variability, correlation and regression) and Inferential Statistics (t-Test, F-Test, ANOVA, etc.) • Analyzing Qualitative Data ○ Theme Generation
1. Identify the ethical issues in research in IT 2. Explain the concepts of privacy, anonymity, confidentiality, and plagiarism to ethically sound research	**Unit Five: Ethical Considerations (2 hours)** • Ethical Principles and Intellectual Property • Ethical considerations in technology research Intellectual property rights • Privacy, Anonymity, Confidentiality, Plagiarism, etc.
1. Develop a preliminary research proposal and write reports 2. Present the final report	**Unit Six: Writing Research Proposals and Reports (3 hours)** • Structuring research proposals and reports in technology studies • Academic writing style and citation techniques for computing research

Major Teaching and Learning Strategy

- Lectures, presentations, group-work, reflections, demonstrations, discussions, discovery, inquiry and question-answer sessions, etc.
- **Direction to Teaching the Course**
  - Discussion on the core ideas of the course
  - Start by exploring the problem and research question(s)
  - Have students write their own research (from the beginning) by following the steps in the course
  - Guide/facilitate students in each unit and support them to develop their own research
  - Students will present their reports

Practical

- Students must be involved in designing their research project from the very beginning. Select a research agenda, formulate the research problem and research questions, review at least five pieces of literature, collect a few sample data, analyze the data by using appropriate methods (qualitative or quantitative), write a final report, and present it. Finally, students must submit the project prior to the deadline.

Assessment Plan

- **In-semester (50)**
  Assignment, written exam, presentation, case study, report, objectives, attendance
- **End-semester (50)**

Question Types	Marks Distribution	Total Question
MCQ	10 Marks	10 Questions
SAQ	24 Marks	7 Questions [Attempt 6 Questions]
LAQ	16 Marks	3 Questions [Attempt 2 Questions]

Marking Scheme for Final Examination:

- Section A (**MCQ**): 10 Questions × 1 marks
- Section B (**Short Answer Question-SAQ**) – Total 7 Questions × 4 marks (Attempt any Six)
- Section C (**Long Answer Question-LAQ**) – Total 3 Questions × 8 marks (Attempt any Two)

Learning Units	Total Credit Hours	MCQs	SQs	LQs	Remarks
Unit One: Introduction to Research	4	2	1	1	One case study related question can be asked
Unit Two: Conducting Review of Literature	4	1	1		
Unit Three: Research Design	13	3	2	1	
Unit Four: Analyzing Data	4	2	1	1	
Unit Five: Ethical Considerations	2	1	1		

Unit Six: Writing Research Proposals and Reports	3	1	1		
Total	30	10	7	3	

References:

- Creswell, J. W. (2014). *A concise introduction to mixed methods research*. SAGE publications.
- Oates, B. J., Griffiths, M., & McLean, R. (2022). *Researching information systems and computing*. Sage.
- Practical Research: Planning and Design 12th Edition by Paul Leedy, Jeanne Ormrod
- Evaluation Scheme